Chicago Food Inspections Data Visualization and Analysis

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1. Introduction

Food safety regulations play an essential role in public health, as they define the guidelines for an industry that every person relies on. According to estimates from the Census Bureau, Chicago, Illinois has the third largest population in the United States at 2.5 million in 2020. In large cities such as Chicago, upholding food safety standards requires canvassing thousands of businesses that are forced to maintain a balance between customer service and proper food handling.

This analysis is guided by a central research question – *Do high-risk geographic clusters show meaningful improvement over time, or do they remain consistent sources of elevated violations?* Food inspection data from the City of Chicago Data Portal collected from 2010 to 2026 was cleaned, visualized, and analyzed to answer this question with attention to risk classification and temporal trends.

2. Data Description

The food inspections dataset from the City of Chicago Data Portal is a collection of records from the Chicago Department of Public Health's Food Protection Program, using standard procedures. The original dataset consists of around 310,000 rows, with 17 distinct columns:

'*Inspection ID'*: Unique ID for each inspection (numeric).

'*DBA Name'*: "Doing Business As" name for each business (text string).

'*AKA Name'*: "Also Known As" name for each business (text string).

'*License #*': Unique license number associated with each business (numeric).

'*Facility Type'*: Descriptor for business specialty (text string).

'*Risk'*: Categorical descriptor of risk level (Level 1 highest, Level 3 lowest) (text string).

'*Address'*: Street address for business inspected (text string).

'*City'*: City of business inspected (text string).

'*State*': State of business inspected (text string).

'*Zip*': Zip code of the address for business inspected (numeric).

'*Inspection Date*': Date of inspection (datetime object).

‘*Inspection Type*’: Reason for inspection (text string).

‘*Results*’: Binary result value, either ‘Pass’ or ‘Fail’ (text string).

‘*Violations*’: Recorded violations for each inspection (text string).

‘*Latitude*’: Latitude of business address (numeric).

‘*Longitude*’: Longitude of business address (numeric).

‘*Location*’: Combination of Longitude and Latitude of business address (numeric).

Each row represents a separate inspection, so the ‘Violations’ column ranges in length depending on the number of recorded violations. This dataset shows that businesses can pass inspections both with and without violations, and that there is no limit to the number of violations that a single inspection can result in. These variables allow for analysis on an operational and geographical level, via risk and violation as well as coordinate and relative locations (zip, address).

A couple of notes from the City of Chicago Data Portal are as follows: A change was made on 7/1/2018 to food inspection procedures that affect the data in the set, and although preprocessed prior to being uploaded, some duplicate rows may exist.

3. Data Preparation

3.1 Data Cleaning

The raw dataset has over 90,000 missing values spread across almost every column. It is important to address these values to avoid issues with visualization and achieving accurate analyses. However, there is also an important dimension to this data in that missing values can provide insight to the analysis. For example, a missing value in the ‘Violations’ column could mean that no violations were observed, that an establishment had closed, or that there simply was an error in the data collection. So, to maintain as much original accuracy while cleaning such a large dataset, a lot of time and attention was spent on the cleaning process.

Null values in each column were treated separately and methodically. Some columns, like ‘AKA Name’ and ‘License #’ were easily filled by finding other inspections for the same location somewhere else in the dataset, and filling with the data from that row. To fill the ‘City’ and ‘State’ columns, we first removed any rows that could not be identified clearly as being in Chicago, Illinois. This dataset contained inspection reports not only from Illinois, but surrounding states like Wisconsin and Indiana, as well as distant states like California. Formatting errors were addressed in the ‘City’ column to clearly identify the city of each inspection. These distant points would conflict with plotting and the scope of this analysis. The remaining null values were filled by using the ‘Zip’ code column when available, or by finding repeat inspections with non-missing values. To address inaccuracies in the ‘Violations’ column, failed inspections with missing violation reports were entirely removed, as they only composed of about 1% of the data. Similarly, rows with missing coordinate values composed only 0.3% of the data, so they were also removed.

The ‘Facility Type’ column was a unique case, as this variable seemed to be listed with subjective categorization. This made implicit assignments difficult, so a filtering method was implemented based on common words that occurred in each facility type. For example, finding the word ‘diner’ or ‘café’ in the business title (‘DBA Name’) of an establishment, reveals locations that are restaurants, over other facility types like “Grocery Store”. This method was used to categorize establishments under “Children’s Services Facility”, “Convenience Store”, “Grocery Store”, “Restaurant”, or “Pharmacy”. These five categories composed almost 90% of the entire dataset, so the remaining missing values were simply set to “Unknown”.

3.2 Data Grouping

In its original form, the food inspections dataset provided little insight. First, it would be necessary to separate the ‘Violations’ column into new rows for each recorded violation, with all of the other columns the same. This increased the size of the dataset from 300,000 to over 1 million rows, which only increases the difficulty of analyzing the data in its original form. As is, the dataset provides reports for qualitative aspects of businesses like their facility type and nickname, as well as insights into specific inspections like the cause of an inspection and the categorized risk level of a location. Inspecting the distinct values of these columns reveals that both facility type and cause for inspection can range in precision, with 363 and 86 unique values, respectively. This makes these variables almost impossible to analyze in a meaningful capacity, since analyzing the original data risks undergeneralizing, while grouping records risks overgeneralizing.

Rather than investigating how these qualitative contribute to risk, it would be more beneficial to create quantitative variables based on inspection outcome. Because the interest of the analysis is the relationship between risk and geography, aggregation was performed on the dataset to group data by Zip code and Risk level with variables:

‘*inspection_count*’: Number of inspections for each risk level in each zip code (numeric).

‘*number_of_failures*’: Number of failures for each risk level in each zip code (numeric).

‘*failure_rate*’: Number of failures divided by number of total inspections (numeric).

‘*number_of_violations*’: Number of violations across all inspections (numeric).

‘*total_violations_at_failure*’: Number of violations only at recorded failures (numeric).

‘*most_common_violation*’: Mode violation code (numeric, categorical).

‘*risk_proportion*’: Proportion of given risk level in each zip code (numeric).

‘*violations_per_inspection*’: Proportion of total violations and inspections (numeric).

‘*violations_per_fail*’: Violations at failure divided by number of failures (numeric).

‘Risk Code’: Risk level represented as integer level (numeric, categorical).

This reprioritized dataset gives insight into geographical statistics, rather than individual records of inspections. These insights provide the foundation of the analysis and visualizations.

4. Exploratory Visualizations

To understand how risk is related to time, it is crucial to establish how risk classification is associated with other variables in the data. For instance, one may assume that high-risk inspections have a higher rate of failure than their medium- and low-risk counterparts. We can test this by plotting the distributions of failure rates across all three risk levels with boxplots.

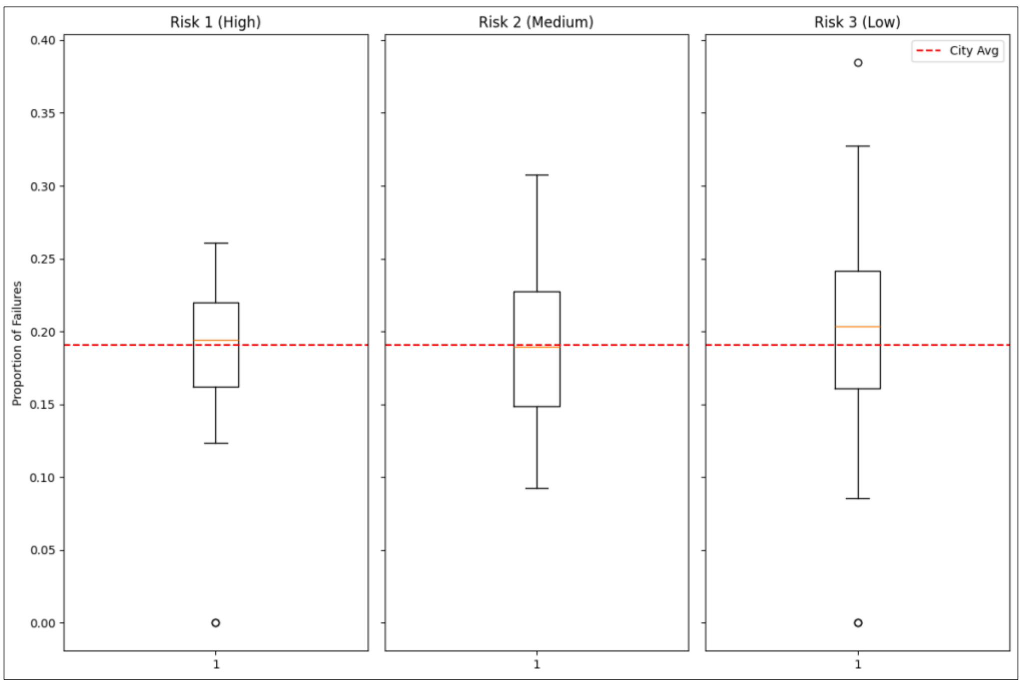

Figure 4. 1

The boxplots in figure 4.1 show that each risk group hovers around the city average, around 0.18. What is surprising about this plot is the decrease in range as risk increases. This could show that there are more high-risk establishments in general. Since failure rate did not provide insight into risk classification, the same process of creating boxplots can be repeated for violations per failure.

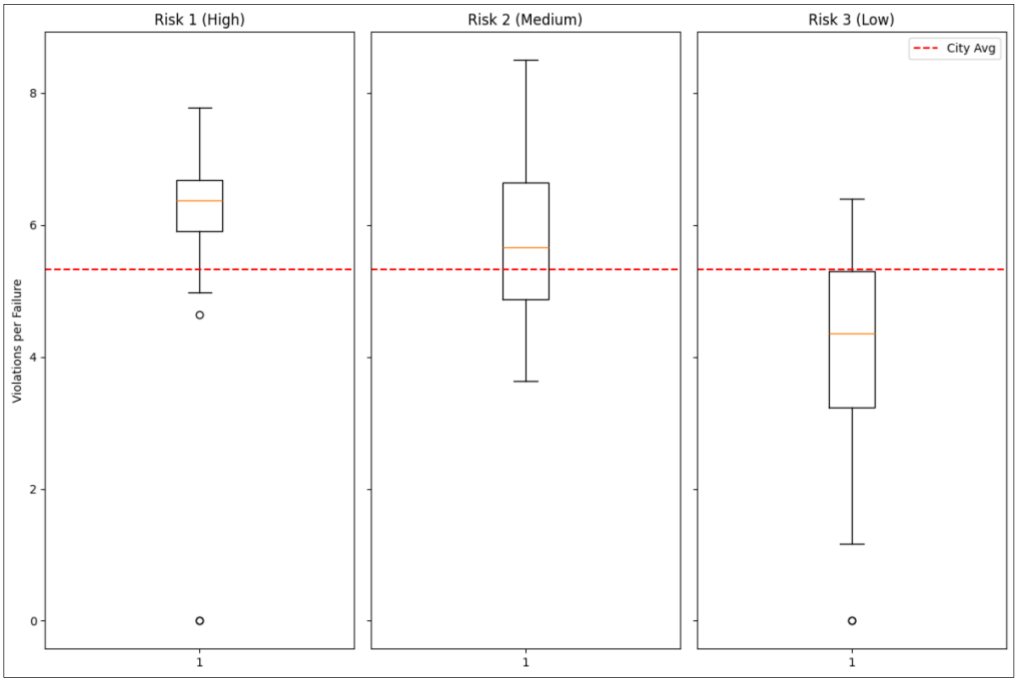

Figure 4. 2

The boxplots in figure 4.2 show a much clearer relationship. As risk increases, the median violation per failure also increases between risk group, to the point where high-risk inspections have a much higher median than the city average.

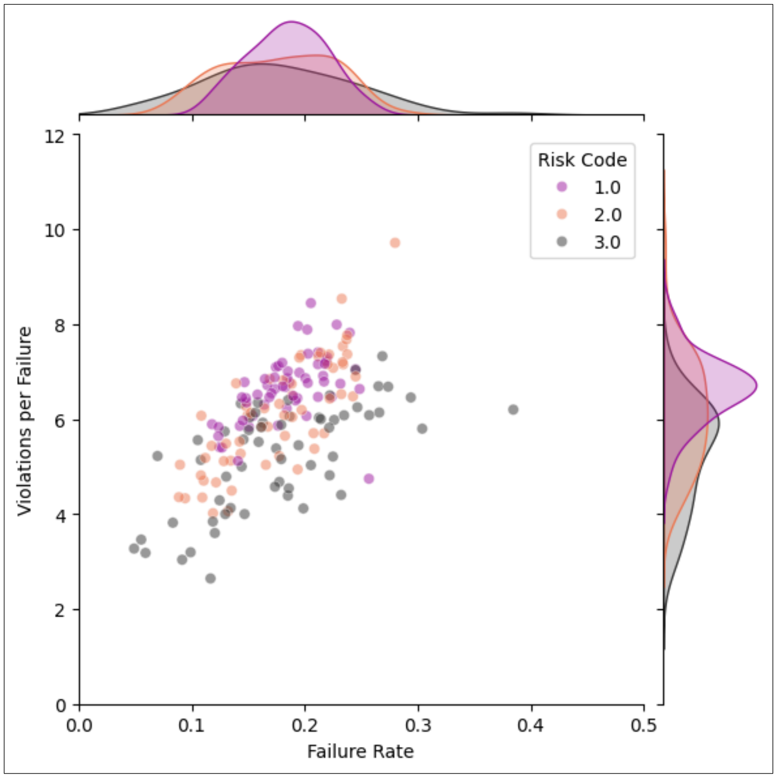

Figure 4. 3

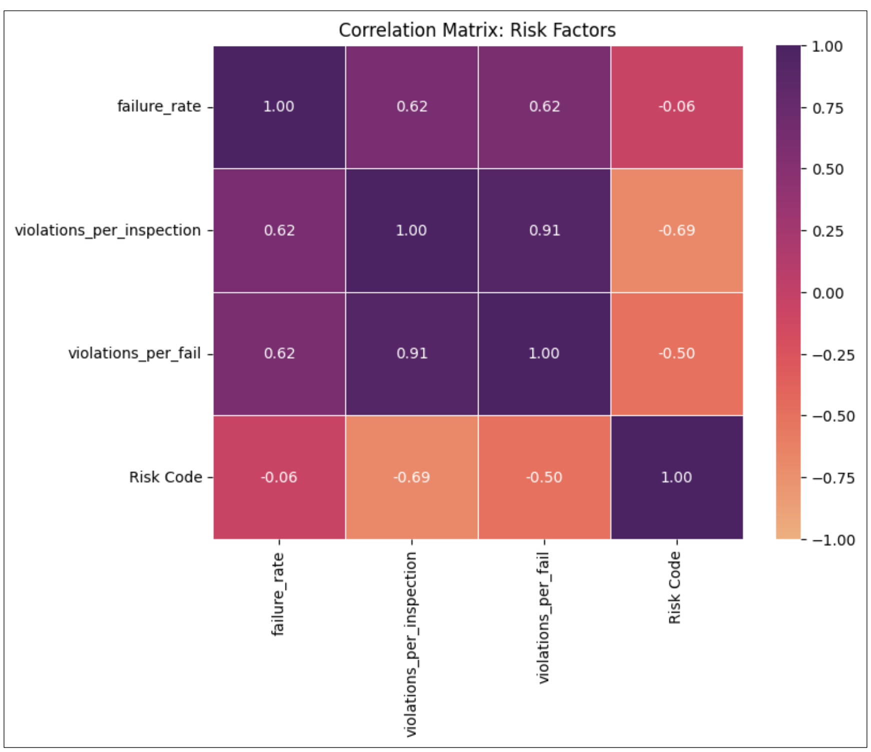

Figure 4. 4

The jointplot in figure 4.3 shows that failure rate is spread mostly between 10% and 30%, which reinforces the findings from the boxplots. However, it also shows that Risk 1 locations, encoded in grey, fall mostly between two and five violations per failure, while Risk 2 and 3, encoded in orange and purple respectively, rarely drop below five violations per failure and often reach up to eight. This discovery is echoed in the density curves on the side of the plot, which show the density of violations per failure for Risk 3 spike sharply around 7, while the other two groups have a much duller, sustained curve. Although Risk 2 and 3 show wider upper ranges in the plot, the overall median violations_per_fail still increases as establishments move toward higher-risk categories.

These associations are further validated by the correlation matrix in 4.4. The plot shows a strong, negative correlation between both violations per failure and total inspections and risk category and a weak, negligible correlation between failure rate and risk category. The negative correlation between the violation variables represents that higher violations correspond to a lower risk code, which in this data represents higher risk.

5. Analysis and Final Visualizations

5.1 Target Data Selection

To identify geographic areas of interest, zip codes were filtered and chosen by a process of elimination. First, it was important to test the earlier assumption that high-risk inspections are more common in general. Across zip codes, the average proportion of high-risk inspections was 75%, with medium- and low-risk following at 18% and 7% respectively. This reaffirms the assumption and suggests that risk proportion alone is not a good indicator of true high-risk areas. Since violations per failure were confirmed to be associated with risk categorization, high-risk entries were extracted from the aggregated dataset and sorted by violations per fail, in descending order.

The zip code with the highest number of violations per failure recorded just more than 400 inspections, which is about 10% the number of inspections in other zip codes, so its proportions are likely skewed. The second highest zip code by violations per failure had only 72% of high-risk inspections, which is below the city average and therefore not ideal for this analysis. The next two zip codes, 60614 and 60659, fit the focus of this analysis much closer, with risk proportions of about 80% and 78% respectively. The next zip code, despite a high number of violations per failure, had a low high-risk proportion at only 62%. The next zip code, 60616, had a high-risk proportion of 78%. For the sake of readability during the visualization process, the analysis will be kept at three zip codes for now.

5.2 Geospatial Plot

To see how the chosen zip codes relate to the surrounding areas, a map of Chicago zip codes can be plotted and colored based on a scale to compare average violations per failure.

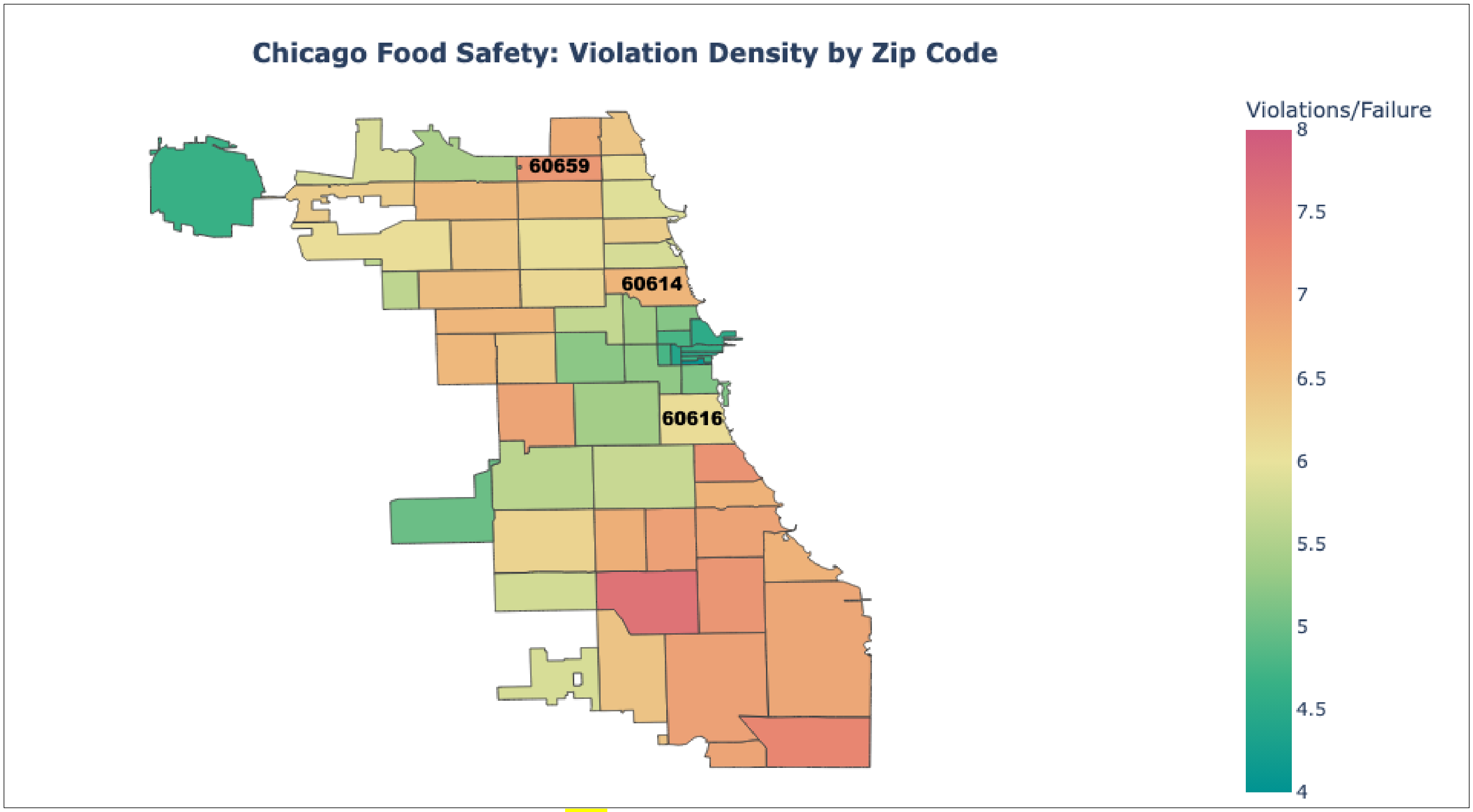

Figure 5.2. 1 – Note:. Find interactive map here: link.

The map in figure 5.2.1 shows that the chosen three zip codes stand out among surrounding areas as potent hotspots for average violations per failure. This map also shows the variation in average violations per failure per zip code. This could be due to missing records of violations, which would heavily weigh areas towards a lower overall violations per failure. This assumption will be tested during time-series visualizations.

5.3 Final Visualizations

To visualize the relationship between violations per failure and time in the three target zip codes, an interactive plot with month on the x-axis and violations per failure on the y-axis was created, with a slider to access each year's data. The full plot can be found at the link here, but for the sake of this analysis, only years with important information will be included in this document.

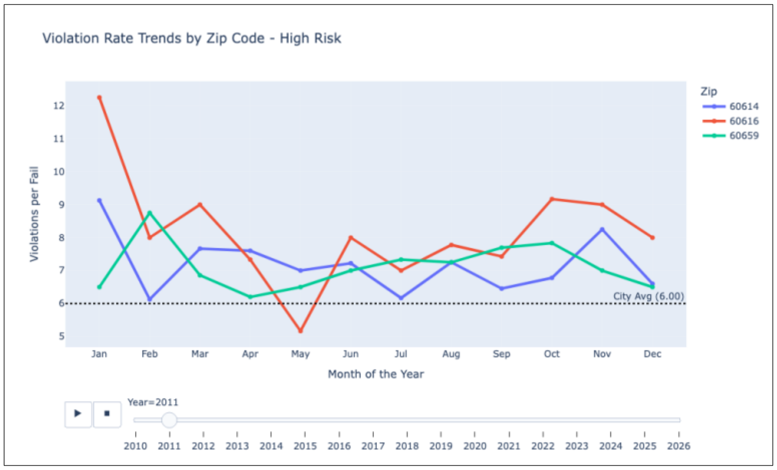

Figure 5.3. 1 - 2011

Figure 5.3. 2 - 2015

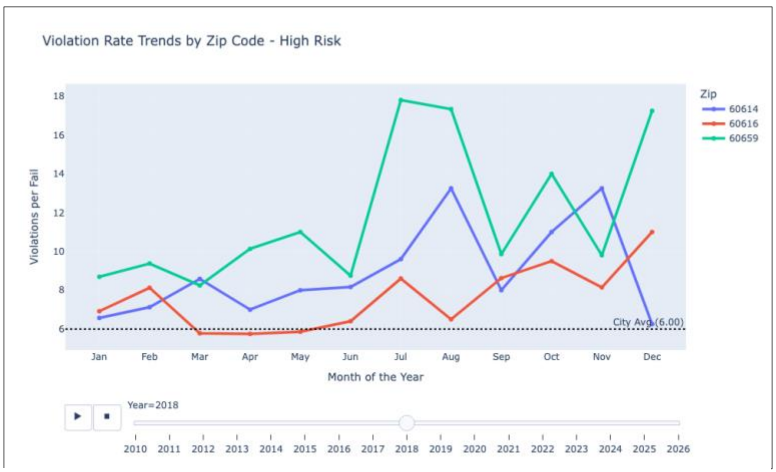

Figure 5.3. 3 - 2018

Figure 5.3.4 - 2020

Figure 5.3. 5 - 2022

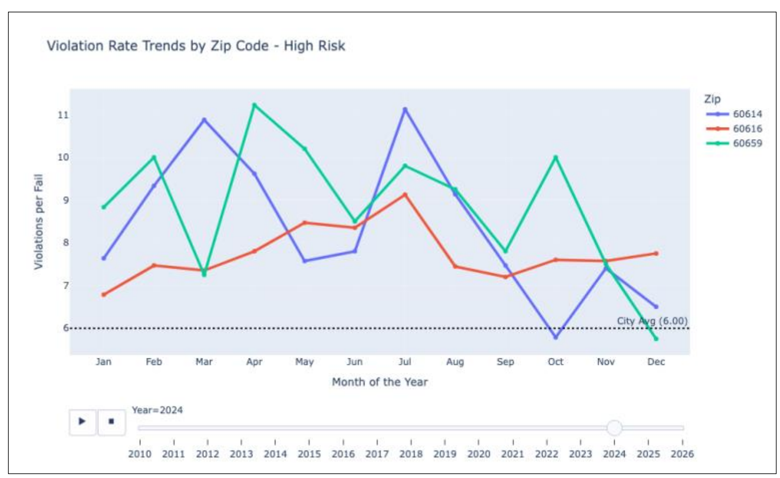

Figure 5.3. 6 - 2024

The time-series plots show a complicated relationship between violations per fail and time. For the first few years, the zip codes appear to hover around the mean and even drop significantly below the mean. In 2018, when inspection procedures were updated, 60616 is the only area that records a month with violations per failure below average. The other two zip codes start and end the year above average, with the 60659-area recording two months over 14 violations per failure, almost ten above the mean. The next year of interest is 2020, when the COVID-19 pandemic broke out across the United States. During the early months of this year, 60614 and 60659 record extremely high violations per failure, with each recording at least one month above 12 violations per failure, which is more than double the city average. 60616 records two months with zero violations per failure, but the high number of violations per failure in the other months suggest that this is likely the result of missing data. These trends seem to carry on to following years, as the plots for 2022 and 2024 show, with 2024 showing that each zip code recorded at least one month above eight violations per failure. The plots suggest that a combination of the procedural changes introduced in 2018 and the COVID-19 pandemic that followed two years later play a significant role in the results of food inspections, and that these effects have contributed to the way that businesses are categorized by risk level.

After seeing how these areas perform monthly during each specific year, it would be beneficial to visualize yearly results to better understand how months of poor performance contribute to ongoing safety hazards.

Figure 5.3. 7- Yearly trends by Mean – find the interactive plot here.

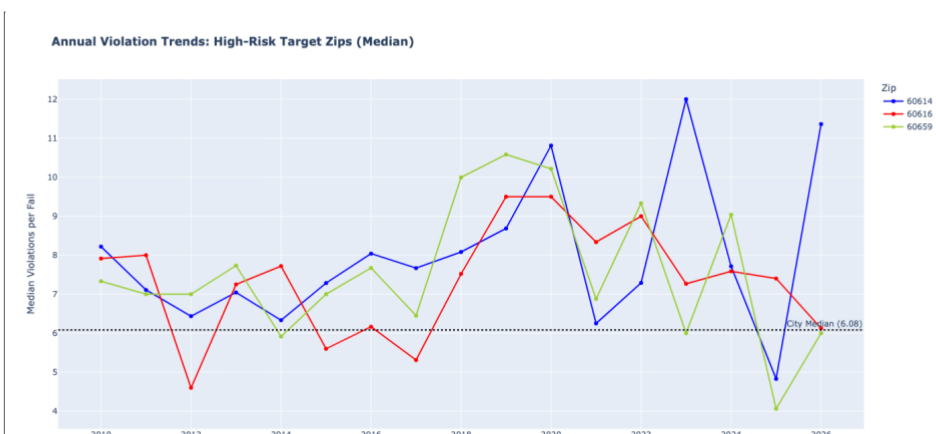

Figure 5.3. 8 – Yearly trends by Median – find the interactive plot here.

The plot in Figure 5.3.7 shows the mean violations per failure for each zip code at each year. This plot reinforces the findings from the monthly plot, as it shows a sharp increase in violations per failure across all three zip codes starting in 2018, and average violations per failure above average for most of the following years. When plotted using the median for each year instead of the mean, as in Figure 5.3.8, the results are even more clear than the first plot. Using this method, the reference for violations per fail is lowered from 5.86 to 5.66, and the violations per failure for the 60614- and 60659-areas increases during earlier years, which suggests that the changes in inspection procedure are not as strong as originally suggested.

For the areas of interest, there is clearly instability in food safety standards between 2018 and 2024. These are the areas that were specifically chosen as being highest-risk, and trends may be different citywide. To answer this, Figure 5.3.9 shows yearly median violations per failure for each risk group plotted from 2010 to the beginning of 2026 along with a citywide median.

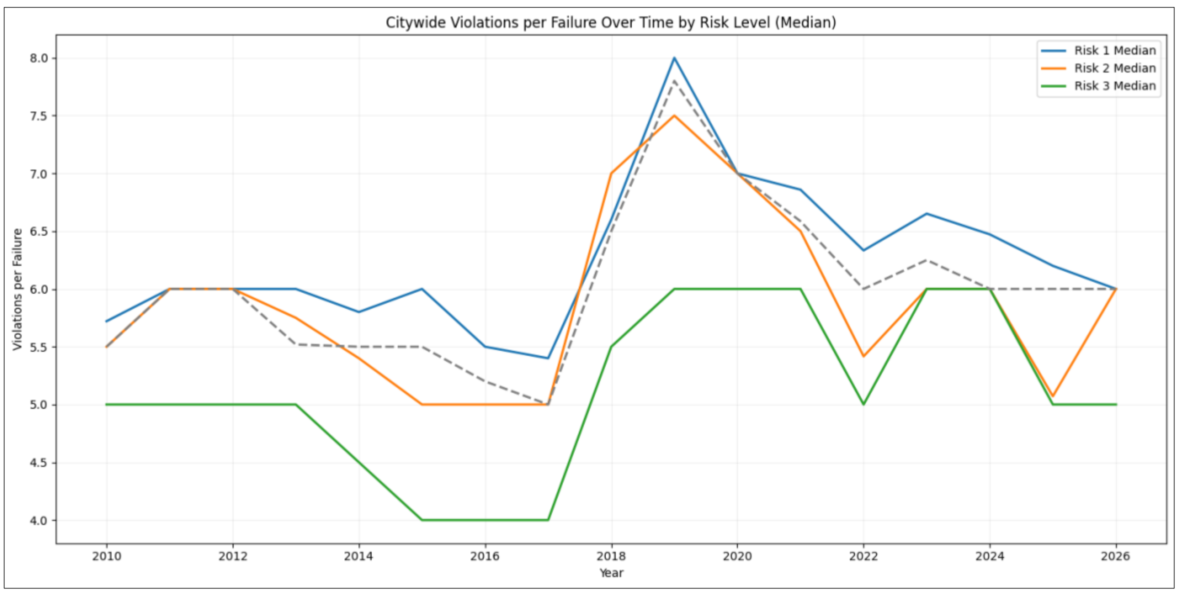

Figure 5.3. 9

While citywide median violations per failure across high-risk inspections from 2018 to 2024 are significantly lower than the ones in Figure 5.3.8, high-risk inspections report more violations per failure than the lower two groups, except in 2018. Median was chosen in this plot to show the true difference between groups, and this highlights a couple of interesting findings. For the first three years in the dataset, Risk 1 and 2 recorded very similar median violations per fail. This could suggest that medium-risk inspections were just as likely to be reported as high-risk ones during these years. After that, risk groups even out and high-risk inspections report more violations per failure until 2018. In 2019, violations per failure peak for each risk group, and surprisingly decrease in 2020. This could suggest a lack of data due to business closures or other irregularities related to the COVID-19 pandemic.

Overall, it appears that rather than the outcome of food inspections changing, the patterns of record keeping themselves may change. However, it can be shown that risk categorization is associated with violation count, and that the number of violations for high-risk inspections has increased since 2010.

6. Limitations and Ethics

The food inspections dataset unfortunately provided limitations that restricted this analysis from achieving a level of precise assumptions. Potentially the largest restriction on this analysis was the breadth of the original dataset and the subjective variability that existed even after cleaning. The qualitative nature of the variables made the amount of distinct report outcomes impossible to compare. For instance, hundreds of unique inspection types made it difficult to linearly track the progression of a business after they failed an initial inspection. Also, the inclusion of comments within the 'Violations' column meant that some inspections had unique, detailed records of how regulations were violated, and others did not. Since the original dataset contained over 300,000 observations, analyzing these comments would not be humanly possible.

Another large limitation of this study was the procedural change in food inspections that was rolled out in 2018. There is mention of this on the City of Chicago Data Portal website, but the details are hidden behind an expired link, so there is no way to accurately assume what this change was. Furthermore, an increase in recorded violations after 2018 was shown, so this change clearly made a significant impact on the data, which impacts the credibility of this analysis.

Finally, the implications of food inspection results on businesses are sometimes as impactful as they are for customers. Once an establishment is labeled as "high-risk" in a high-population area like Chicago where customers have options, they will choose other businesses. For the sake of food safety and small businesses alike, it is important that organizations in charge of food inspections are consistent and clear about their inspection methodology.

7. Conclusion

In summary, high-risk geographic clusters in Chicago have not shown significant improvement over time, and high-risk areas have only reported more violations since 2018. However, the data also suggests the presence of underlying causes like changes in inspection procedure that could confound with inspection results. In order to produce an analysis worthy of public support, the City of Chicago Data Portal must ensure transparency in both its methodology and its transparency in its procedures.

References

City of Chicago (2026, May 4) *Food Inspections Data*. https://data.cityofchicago.org/Health-Human-Services/Food-Inspections/4ijn-s7e5/about_data

Census Bureau (2020, July 1) *Total population estimate. https://worldpopulationreview.com/us-cities*